Camphene Day 3 – 11/14/2023

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CHEM 2081 – Section 003

Introduction

In this experiment, simple reflux was used to obtain camphor from borneol. The chemicals used in this experiment include glacial acetic acid (CH3COOH), acetone (C3H6O), aqueous sodium hypochloride (NaClO), methylene chloride (CH2Cl2), saturated sodium bicarbonate (CHNaO3), 5% sodium bisulfate (NaHSO4), anhydrous sodium sulfate (Na2SO4) and water (H2O).

Methods and Materials

A 10mL long-neck, round-bottom flask was obtained, and racemic 0.3g of borneol product was added along with 0.3 mL of glacial acetic acid. Initially, 1 mL of acetone was used to dissolve the racemic mixture, and a boiling chip was added. However, if the borneol did not dissolve at a temperature range of 40°C-50°C, an additional 0.5 mL of acetone was incrementally introduced until complete dissolution occurred. Following dissolution, 6 mL of aqueous sodium hypochlorite solution (bleach) was slowly added drop by drop through the top opening of the condenser. After each addition, gentle swirling ensued, and the reaction was allowed to continue for 15 minutes post the final bleach addition. Subsequently, the mixture was allowed to cool to room temperature. The camphor was extracted using two 2mL portions of methylene chloride (DCM) in a centrifuge tube, isolating the organic layer, which surfaced as the top layer.

The methylene chloride layers were neutralized with approximately 2 mL of saturated sodium bicarbonate while checking the pH along the way. Additional sodium bicarbonate was added as needed, indicated by the production of CO2 bubbles after each addition of the bicarbonate solution. Sequential washing of the organic layer followed, first with 2 mL of 5%

sodium bisulfate and then with 2 mL of water. To ensure dryness, the organic solution was dried using anhydrous sodium sulfate. The solution was considered dry when freely rolling crystals of the drying agent were observed. The next step involved evaporating the DCM on a watch glass over a water bath, ensuring the temperature did not exceed 50°C. Crystals were collected and weighed to determine the percentage yield of the camphor obtained.

Results

1. The sample on the watch glass from last week was still an opaque, white paste instead of free rolling powder or crystals. It is possible that the sample did not dry properly at the end of the second day of the experiment, or throughout the week. There was about 0.217g of this paste. About 0.087g of a provided sample of the day 2 product was added to this sample to have 0.304g of sample in total.
2. The syringe to collect the 0.3 mL of glacial acetic acid was not properly marked, may not have been a precise measurement of this acid.
3. The water bath was set at about 45°C.
4. After adding the bleach and swirling the solution, it became cloudy.
5. When adding the DCM, first, 2 mL were added. Two layers formed after this addition, the organic solution was washed, but it was hard to distinguish where each of the two layers ended. Added 2 mL more of DCM, washed the organic layer, then separated the organic (top) layer. When extracting the DCM (bottom) layer, the boiling chip was stuck in the bottom and was not able to be removed with the pipette. This may have caused some DCM to stay with the organic layer.

6. After adding 2 mL of saturated sodium bicarbonate, the pH of the solution was checked with red litmus paper. The dot on the litmus paper did not turn blue, so it was presumed that the solution was properly neutralized.
7. It was stated in lab that, when adding anhydrous sodium sulfate, a spatula tip per mL of solution should be added. Roughly 5 spatula tips, or 0.378g of sodium sulfate was added. The solution fizzed after each addition of the sodium sulfate, then the bubbles calmed.
8. The sand bath started at 50°C, but to speed up the process, it was increased to 60°C.
9. It took a very long time to dry the organic layer solution over the water bath. After about 30 minutes of waiting for it to dry, only for it still to be a completely liquid solution, it was determined that along the way water had been collected instead of the intended organic product. It is possible that there was a drying agent issue involved.

Calculations

- No percent yield calculation can be completed because no final product was collected.

Discussion

In this experiment, simple reflux was used to obtain camphor from borneol. Quantitative assessment of the refluxed mixture suggests that an aqueous solution more closely resembling water was collected, rather than the anticipated product camphor. After the week-long reaction, there was approximately 0.217g of white borneol paste remaining on the watch glass, rather than a white powder or white crystals. After refluxing the mixture and separating the organic layer from the DCM layer, the liquid organic layer was put onto a watch glass sitting on top of a hot water bath to evaporate the extraneous water from the mixture and form crystals. However, after waiting for almost 40 minutes for the solution to evaporate, the sample remained completely

liquid. It was then determined that the sample on the watch glass was a majority water, and thus a very small amount of product would form, even if there was more time to evaporate the solution.

Possible sources of error include stopping the reaction too early and not giving the reaction enough time to fully complete, in which case a fix for future iterations could be to allow the reaction to run longer. Another source of error may be the use of impure starting material, or improper yield of starting material. If the initial amount of borneol was incorrect or it was impure, then this would affect the overall yield. For future iterations, it may be necessary to redo the drying portion of the Camphor Day 2 experiment to obtain white powder or crystals, rather than a white paste, to then dry for the week to complete the Camphor Day 3 experiment. It also may be necessary to detail a more exact amount of drying agent to be put into the solution before heating the organic layer over a watch glass.

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry II Lab Manual [lab manual]. Auburn (Al): Auburn University.

in watchglass ⇒ possible didn't dry enough ⇒ not white crystals, but clear, dry clump of "paste"
↳ had about 0.217g, added more to get to 0.304g (do math)

Experiment 6 Procedure – Camphene Day 3 – 11/14/23 – Sophia Halstead

1. Obtain a 10mL long-neck, round-bottom flask. Add racemic 0.3g of borneol product, as well as 0.3 mL of glacial acetic acid. Add 1 mL of acetone to dissolve the racemic mixture. Add a boiling chip.
   ↳ syringe x well marked, ~0.3mL
   ⇒ didn't have, got from other people

2. Assemble a reflux with ~~the~~ distillation column. Place the apparatus in a water bath at (45°C) 40°C-50°C. If borneol does not dissolve at this temperature, add 0.5 mL of acetone until all borneol goes into solution.

3. Once all dissolved, add 6 mL of aqueous sodium hypochloride solution (bleach). Add this through the top opening of the condenser and add dropwise ~~over 15 minutes~~ slowly. Gently swirl after each addition. Allow reaction to continue for 15 minutes after the last bleach addition. (soln became cloudy)

4. Let the mixture cool to room temperature. Extract the camphor with two 2mL portions of methylene chloride (DCM). Use a centrifuge tube to extract. The organic layer is the top layer (?).
   ⇒ added 2 mL, 2 layers slightly formed, washed, still hard to see end / beginning of layers. Added 2 more mL, washed, seperated layers.
   (Boil chip still stuck in bottom, may have kept some DCM w/ organic layer)

5. Neutralize methylene chloride layers with ~2 mL of saturated sodium bicarbonate. Add more if necessary. () The presence of acidic residuals will be determined by the production of CO_2 bubbles after every addition of the bicarbonate solution.
   ⇒ if keeps bubbling ↓ (check pH along the way)
   (small bubbles visible)
   (checked pH w/ wire of litmus paper, seemed to be neutralized)

6. () Wash the organic solution sequentially with 2 mL of 5% sodium bisulfate and with 2 mL of water.
   ↳ mostly red, very slight purple tinge
   ↳ may have been less than 2mL water
   layer
   ~~bottom~~?

7. Dry organic solution with anhydrous sodium sulfate. Considered dry with freely rolling crystals of the drying agent are observed.
   ↳ heat on WG over water bath (SB@ 50 ↳ then to 60)
   ↳ "spatula tip per mL" → 0.378 g
   ↳ ~5 spatula tips
   (fizzed after each addition)
   (still small bubbles)
   ↳ heat over water bath to evap DCM

* replace word "solution" w/ layer, DCM x down drain, calc % yield

8. Evaporate the DCM over a water bath at MAXIMUM 50C. Collect and weigh the crystals. Calculate percentage yield. ~~Compare melting point with range in published literature.~~

TOOK a VERY LONG time to dry over the water bath
↳ after abt 30mins of waiting for it to dry, determ. prod. collected water instead of organic prod, maybe drying agent issue

Error sources from last lab can carry over

Lab Performance

Exp 6 D3

Sophia Halstead	3	402	B
Name:	*Section:*	*Room:*	*Hood:*

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves.	○ ○ ○ ○ ○ (5pts)
Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks.	○ ○ ○ ○ ○ (5pts)
Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts).	○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts)
Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts).	○(1pt) ○ (5pts) ○ (5pts)
Working independently or equal partner participation during the entire lab.	○ ○ ○ ○ ○ (5pts)
Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace.	○ ○ ○ ○ ○ (5pts)
Maintain clean and organized personal space.	○ ○ ○ ○ ○ (5pts)

GTA notes:

GTA signature: _____ *Final Performance Grade:* _____